

4.5 Spline Interpolation [Cubic or third-order, aka Hermite spline]

Cubic splines (section 18.6.5)

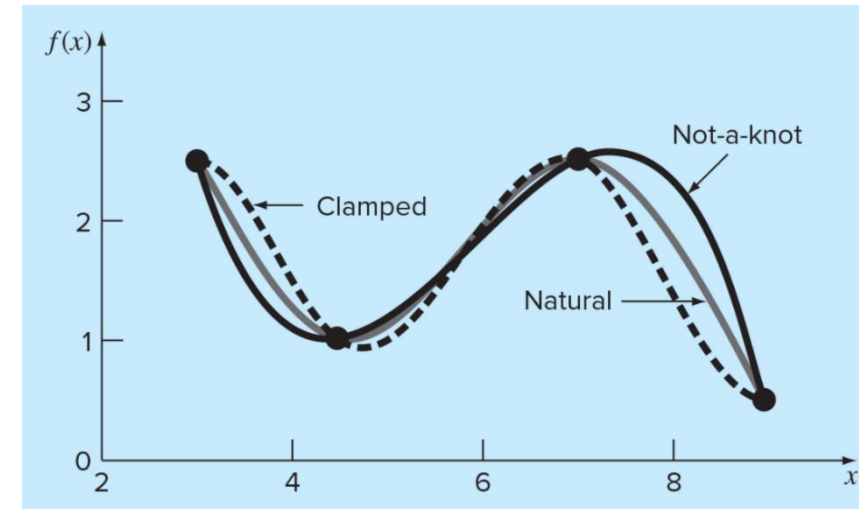
Three commonly used end conditions:

1. Natural spline (second derivatives = 0 at end points)
(straightens at ends due to 2nd derivative = 0)
2. Clamped end condition (first derivatives = 0 at the end points)
(levels off to zero slope at ends)
3. Not-a-knot end condition (force continuity of the third derivative at the second and next-to-last knots).
(non-zero 2nd derivatives, exhibits more curvature)

Notes

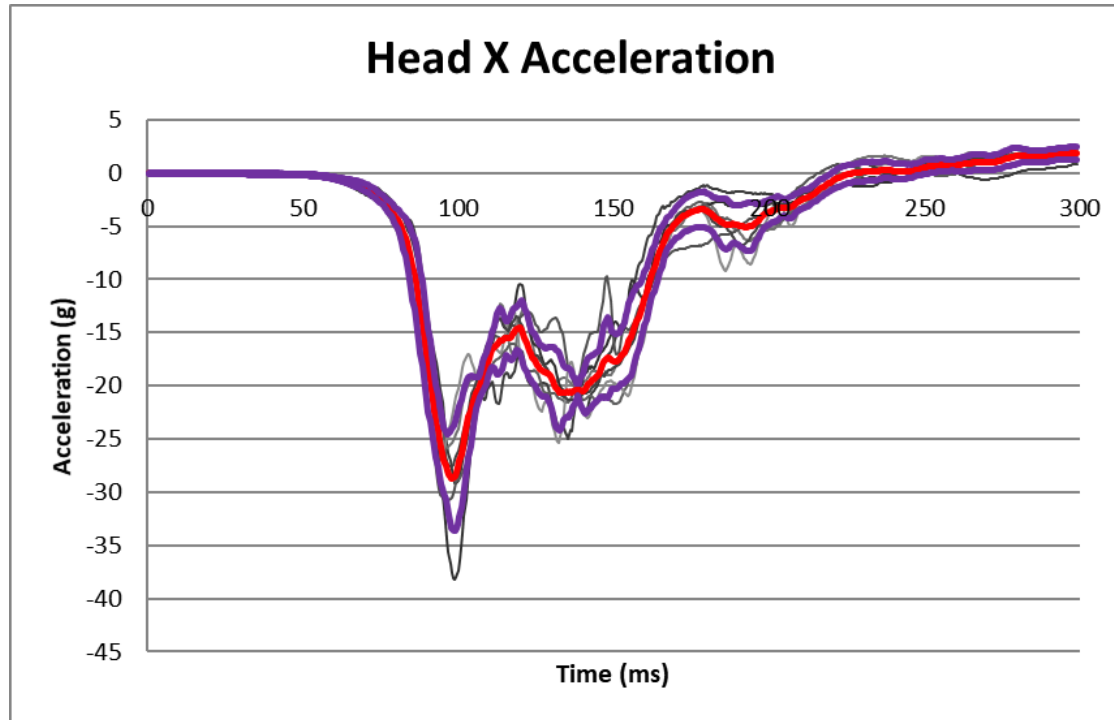
Cubic spline: cubic polynomial with continuity in function values, first derivative, second derivative.
Best for engineering data.

Hermite spline: cubic polynomial with continuity in function values and first derivatives
Commonly used in interpolation of data.
Beneficial if we want to avoid overshoot, or for noisy data.



4.5 Spline Interpolation

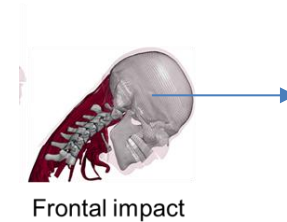
- An example with more data points
- Head x-acceleration (deceleration event) for a frontal sled test



- Data from several human subjects (grey lines)
- Average response (red line)
- +/- 1 standard deviation (purple lines)

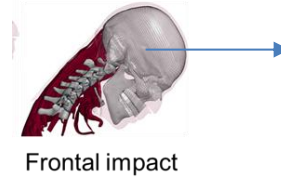


Fig. 6 - Front view of pelvis mount attached to subject

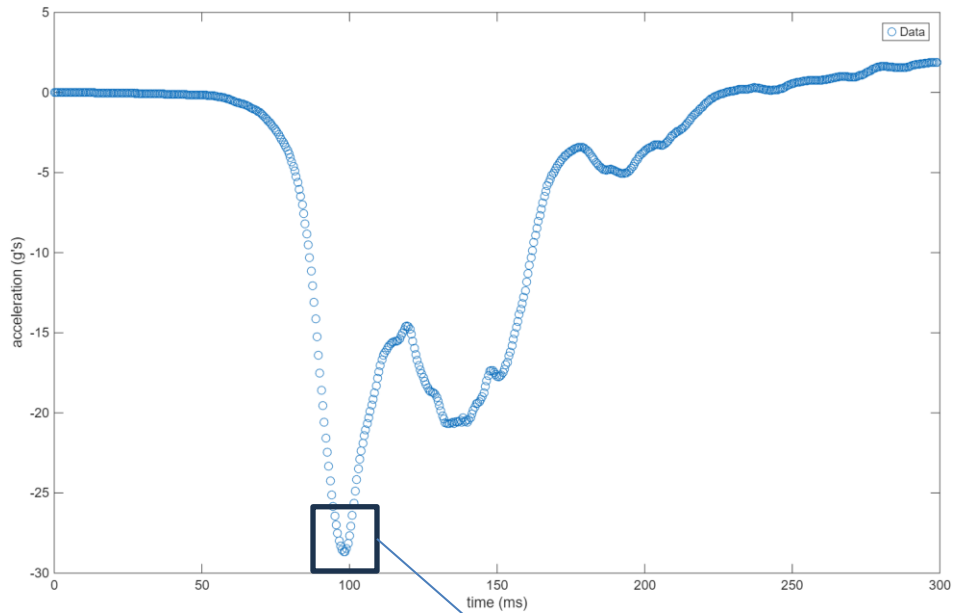


Head CG x-acceleration
(-ve or rearward)

4.5 Spline Interpolation

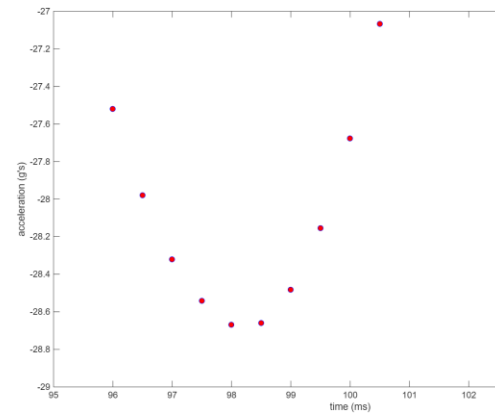


- Average head x-acceleration plotted in MatLab



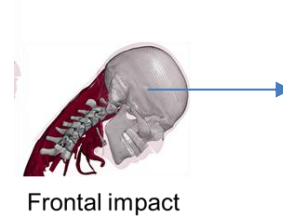
```
1 % xy data from NBDL impact tests
2 % x-axis is time in milliseconds (ms)
3 % y-axis is head acceleration in g's (multiples of gravity)
4 % g = acceleration (m/s^2) / 9.81 (m/s^2) [dimensionless]
5 %
6 x = [0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5 5.5 6 6.5 7 7.5 8 8.5]
7 y = [0 -0.001987 -0.002845714 -0.002429857 -0.005670286 0.00166]
8 %
9 % plot original data (0.5 ms intervals)
10 plot(x,y,'o');
11 xlabel('time (ms)');
12 ylabel('acceleration (g's)');
13 hold on;
14 %
```

File: splinev5nbdlv2.m

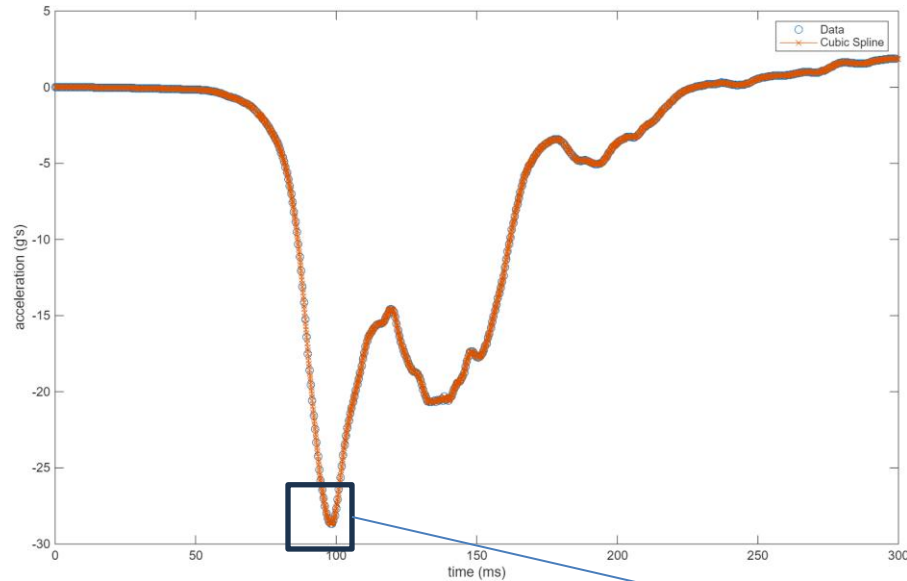


Zoom in on the data at the maximum acceleration magnitude

4.5 Spline Interpolation

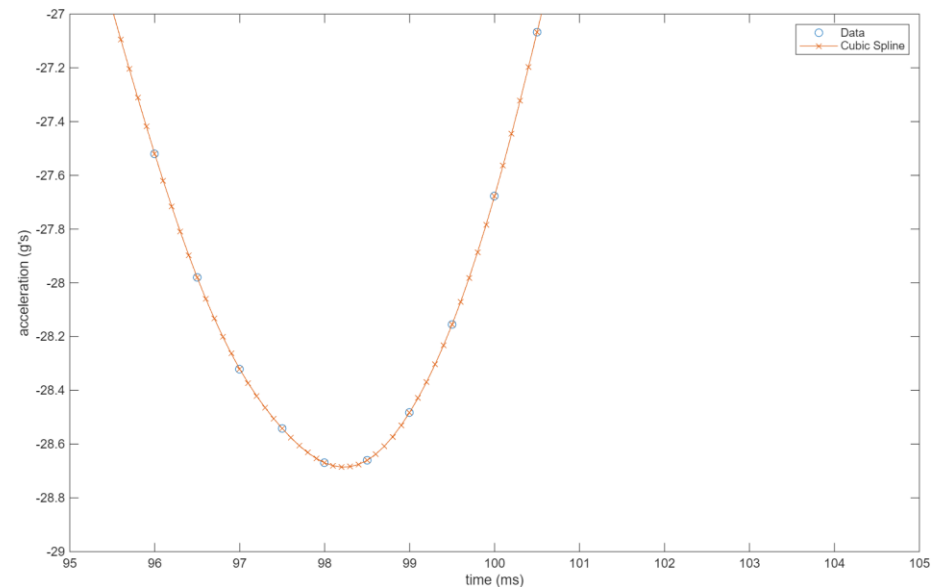


- Average head x-acceleration plotted in MatLab



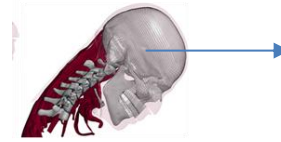
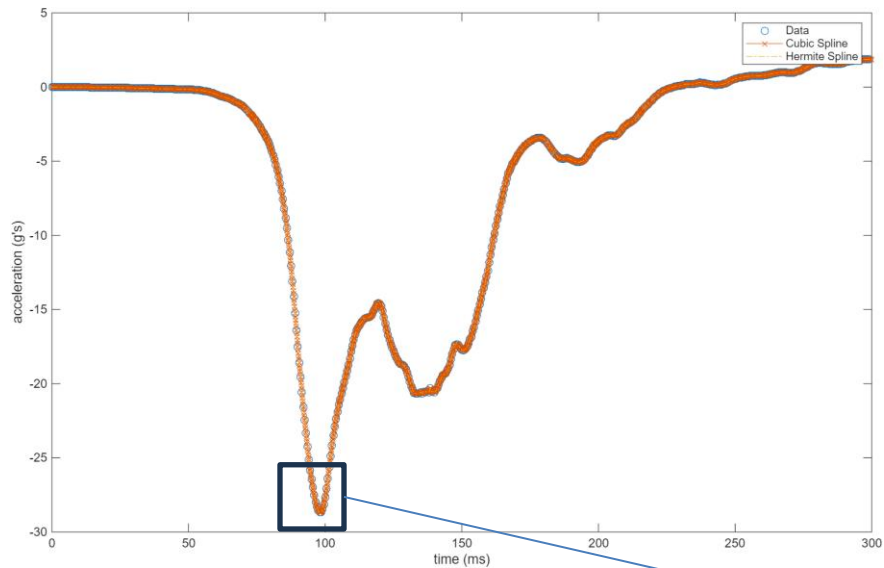
Cubic spline fit to the average curve
Interpolated data points at 0.1 ms intervals
(Original data points at 0.5 ms intervals)

```
15 % generate x-values for spline fits
16 xx = 0:.1:300;
17 %
18 %
19 % Cubic spline interpolation in matlab (continuous in 1st and 2nd
20 % derivatives, smooth but can overshoot- best for engineering data
21 yy = spline(x,y,xx);
22 plot(xx,yy,'-x');
23 hold on;
24 %
```



4.5 Spline Interpolation

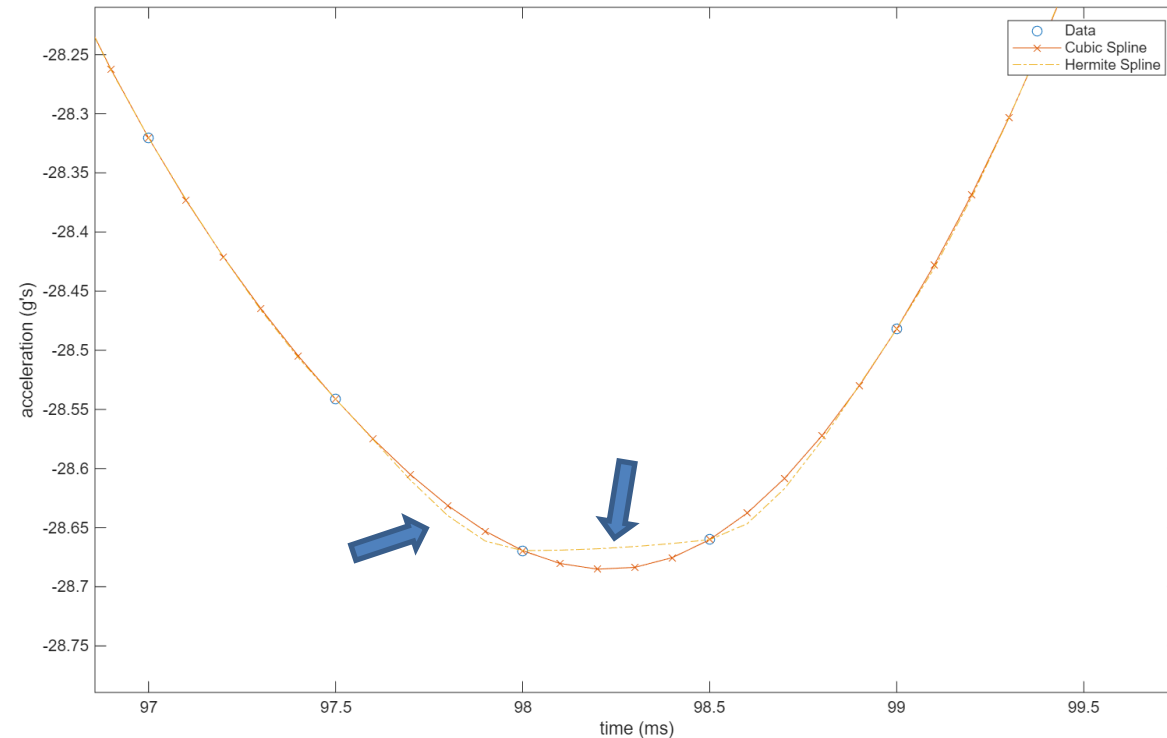
- Average head x-acceleration plotted in MatLab



Frontal impact

Hermite spline fit to the average curve
Interpolated data points at 0.1 ms intervals
(Original data points at 0.5 ms intervals)

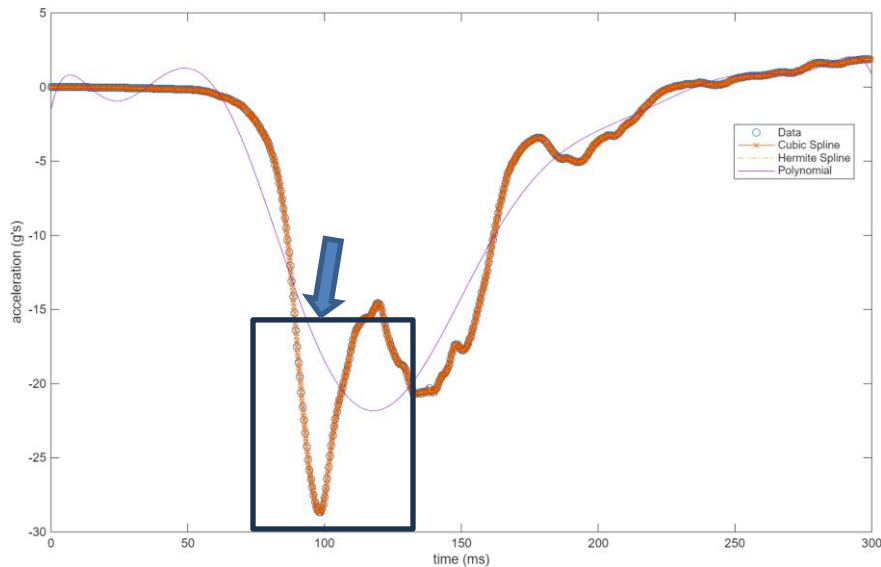
```
30 % Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) - continuous
31 % only in 1st derivative, no overshoots - used for shape preservation
32 p = pchip(x,y)
33 pp = ppval(p,xx);
34 plot(xx,pp,'-');
35 %
```



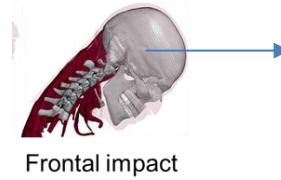
4.5 Spline Interpolation

- Average head x-acceleration plotted in MatLab

10th order polynomial



```
Warning: Polynomial is badly conditioned. Add points with distinct X values, reduce the degree of the polynomial,  
> In polyfit (line 85)  
In splinev5nbd1v2 (line 38)
```



Polynomial fit to the average curve
(** this is a polynomial regression, not interpolation)

Plotting data points at 0.1 ms intervals

```
37 % Polynomial fit to data (last # is order of polynomial)  
38 p=polyfit(x,y,10);  
39 y1 = polyval(p,xx);  
40 plot(xx,y1);  
41 hold on
```

50th order polynomial
(better at fitting major peak but oscillations at ends)

